

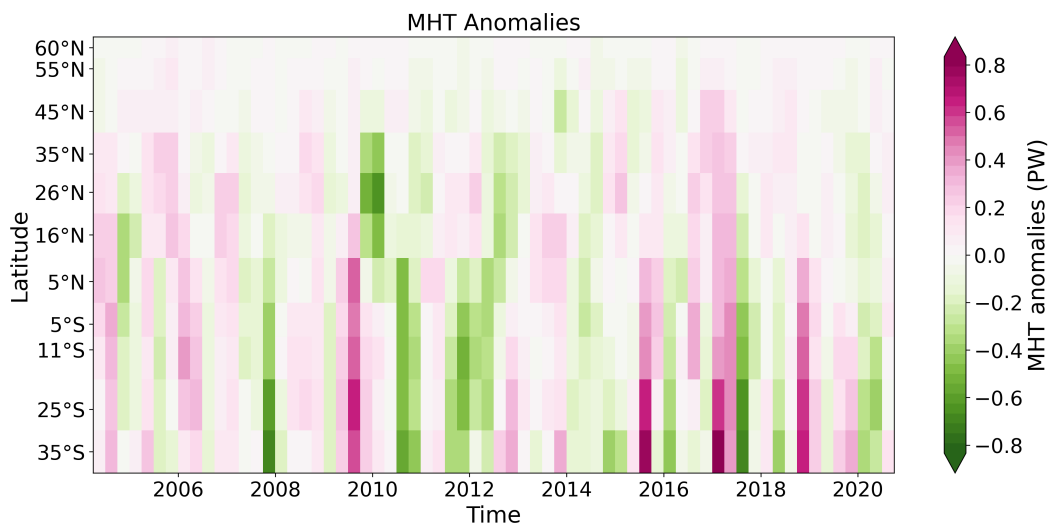
Masterthesis

1. Plot outline

1.1 How coherent are MHT anomalies?

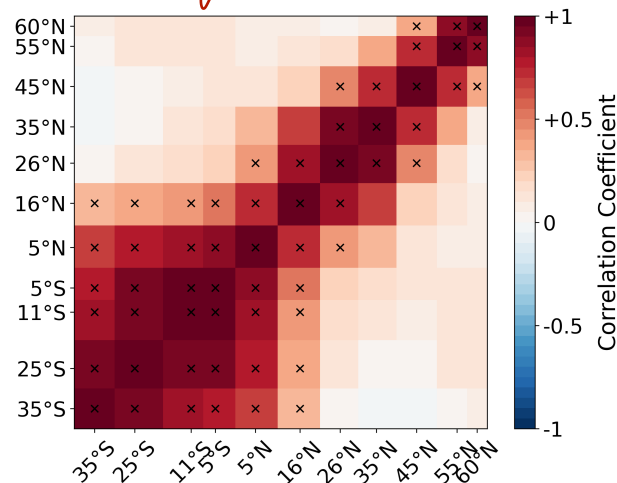
Der Satz sollte alleine stehen können, also auch ohne die Frage darüber.

1. The hovmöller diagram of the meridional heat transport anomalies hint towards latitudes where anomalies vary together, namely between 35°N - 16°N and 5°N - 35°S . Overall weakest anomalies north of 35°N , in subpolar region.



2. The cross-correlation between all latitudes shows that those region have internally high correlation, while the correlation drops between 16°N and 5°N , as well as north of 35°N .

→ der Satz ist noch nicht ganz klar in seiner Aussage. Was genau willst du hervorheben?



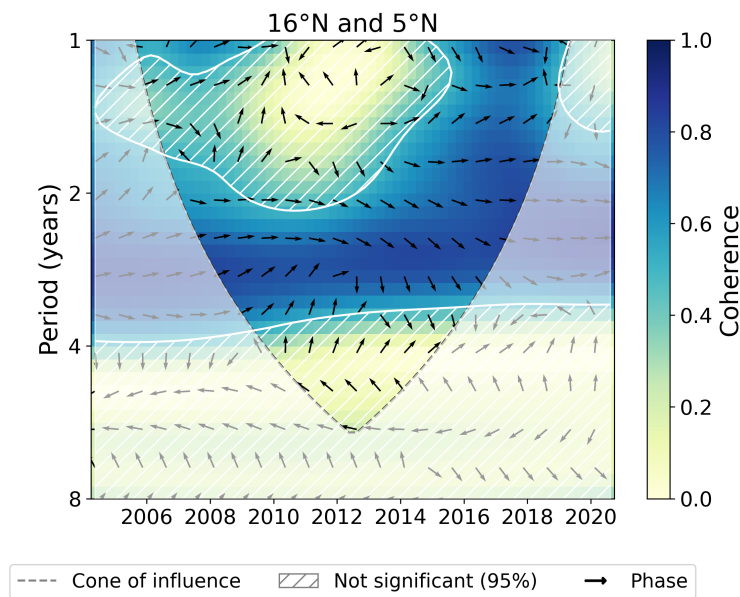
4. As 16°N and 5°N seems to be the border region, where the connection between latitudes changes I will use wavelet coherence to further investigate.

The wavelet coherence between 16 and 5°N reveals a drop in the coherence of the meridional heat transport occurring between 2009-2014. Before 2009, coherence is still not significant but still slightly higher than after 2009.

Warum erwähnst du das? Könnte noch klarer formuliert sein

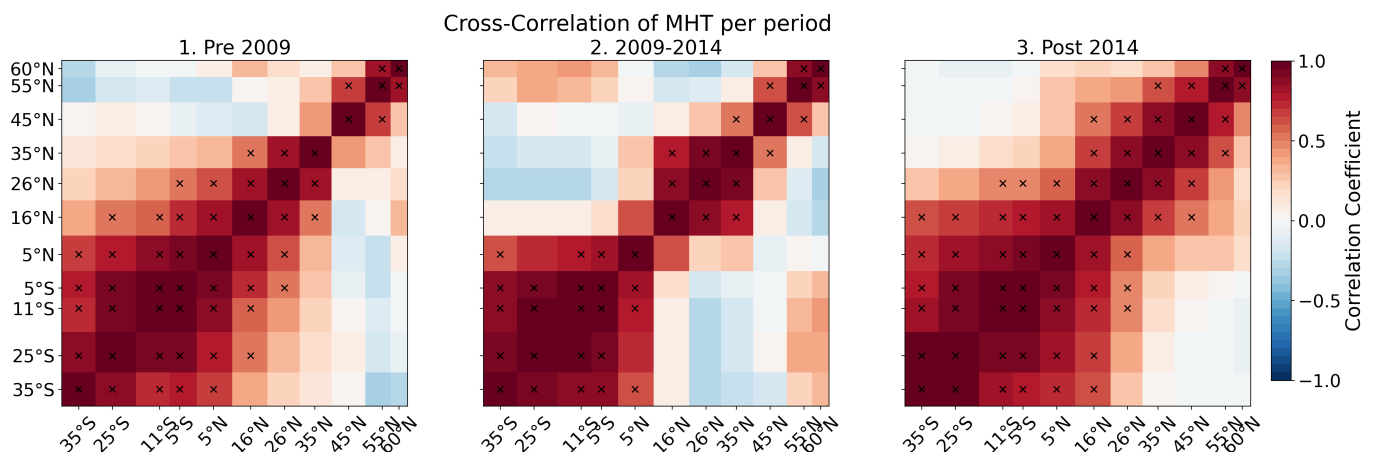
4.1 Three periods can be distinguished: 2004-2008, 2009-2014, 2015-2020

(4.2 for periods around 3 years 16 and 5 stay connected, with some lead lag pattern emerging (2010: 16°N leads 5°N by ~0.5 years & then 2014 vice versa (careful here since time period is very short))



5. MAIN FIGURE

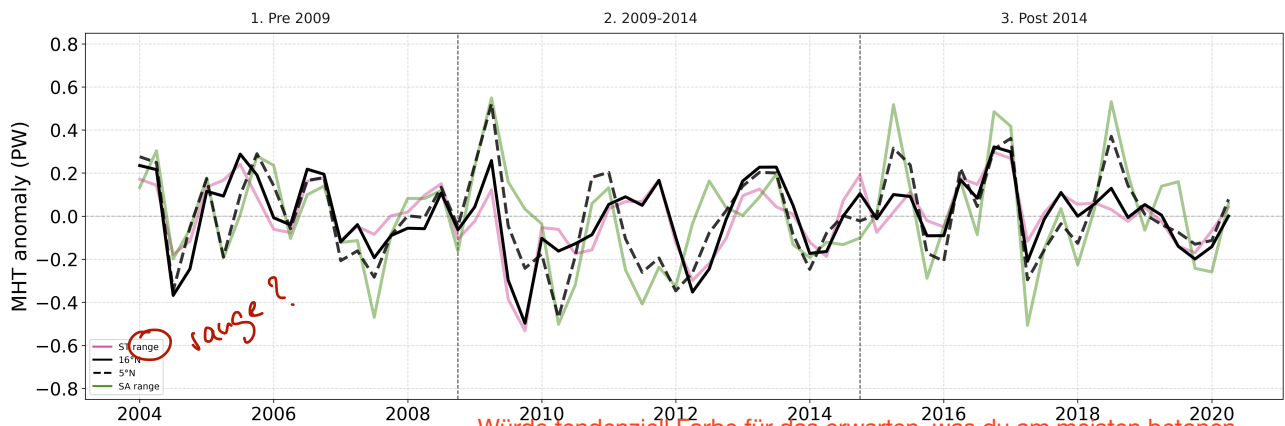
Applying the cross correlation on the MHT anomalies for the three identified periods reveals that prior to 2009 and after 2014 the meridional heat transport seems more strongly coherent throughout the Atlantic south of 35°N, while in 2009-2015 the coherence splits, yielding two boxes of coherent MHT.



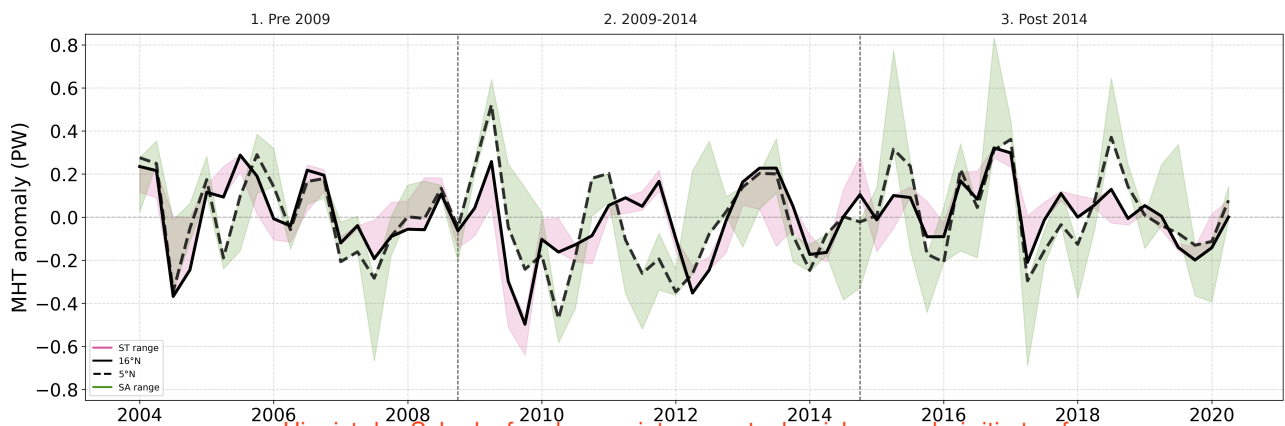
Du könntest noch einen zusammenfassenden Satz zufügen wie: there is a breakdown of coherence in ... der das ganze nochmal auf den Punkt bringt.

2009-2014

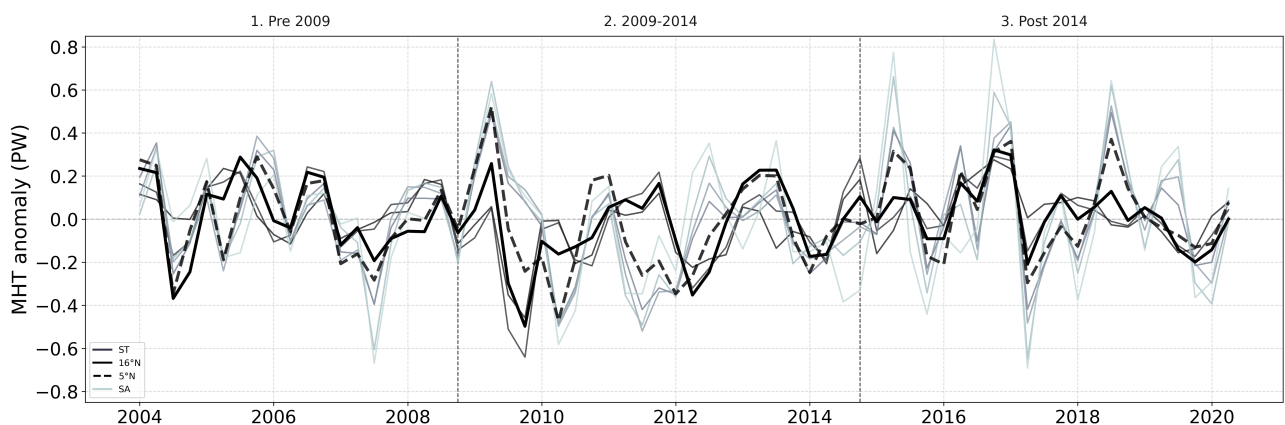
5.1. In the timeseries we can see that especially for 16°N and 5°N in the second period the anomalies are not varying together, even slightly out of phase, where in the 1st and 3rd period they show more in phase variability.



Würde tendenziell Farbe für das erwarten, was du am meisten betonen willst, daher kann man da nochmal nachdenken. Aber sonst finde ich 1. am einfachsten zu verstehen.



Hier ist das Schadenfreude zwar interessant, aber ich war sehr irritiert anfangs, warum es so aussieht als wäre es nur die untere Grenze für die schwarzen Linien, daher vielleicht ungünstig



Remark: not sure which plot is best for the timeseries (16 and 5 always highlighted, thicker black lines)

1. North Atlantic Subtropical latitudes (35-16°N) and South Atlantic (5°N-35°S) as mean
2. Blocks as min max shading
3. Individual lines, with dark gray for northern ones and light gray for southern ones

Siehe oben, aber kannst du Eleanor auch schicken so

6. The average correlation between the two blocks drops for the 2nd period (2009-2014) from 0.53 to a no longer significant value (0.02). After 2014 it increases again to 0.47, while slightly missing the significance threshold of 0.48 (95%).

Plot noch nicht da?

Remark: if I would use 90% for the significance threshold everywhere, it doesn't change any of the main results, however the correlation reached after 2014 would be significant. Should I still stick to 95%, but showing the threshold of 0.48?

7. The north Atlantic subtropical and south Atlantic blocks remain internally coherent throughout the complete time period (2004-2020).

Remark: should I leave the 0-lag figures as a result here or in the appendix?

plot einfügen?

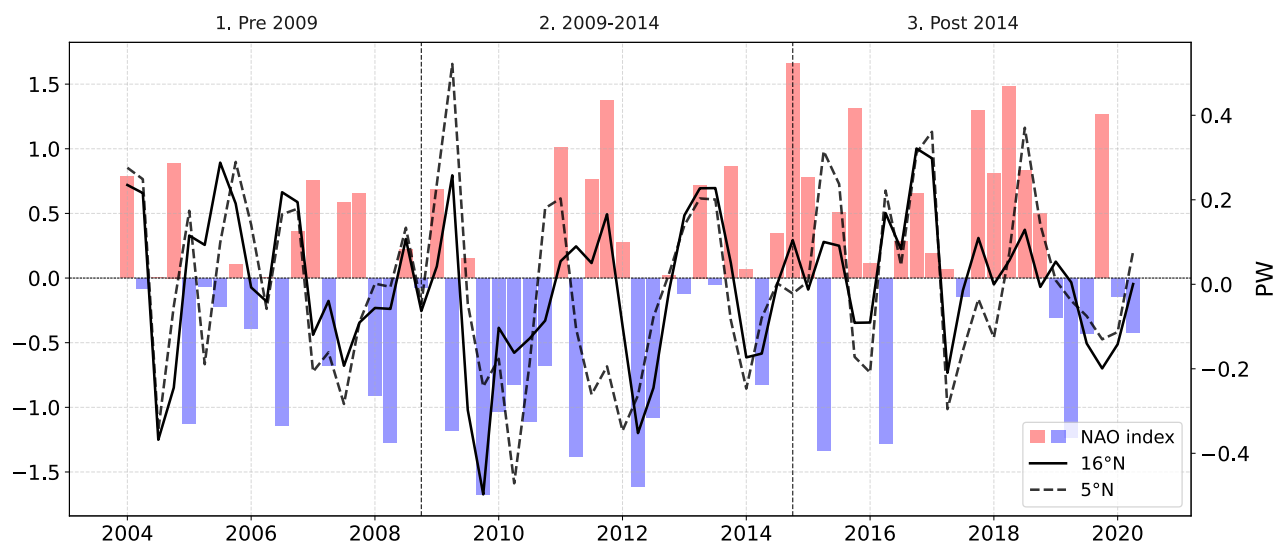
1.2 Why does the coherence break at 16 and 5°?

Now that I established a change in pattern of coherence in a distinct time period (2009-2014) between 16°N and 5°N I will look at possible drivers and changes in the overall heat budget of the region.

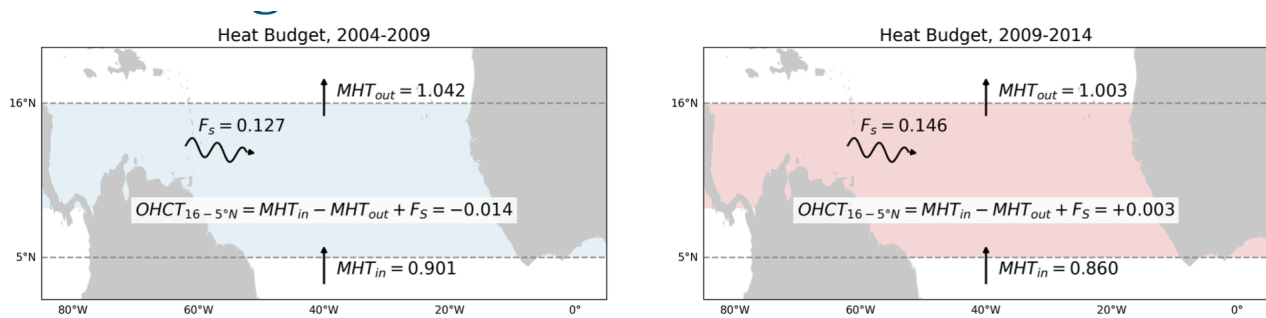
9. in winter 2009 and spring 2010, there as anomalous negative NAO. Correlation between subtropical NA latitudes (35, 26, 16) and NAO index is for the period following this event (2009-2014) around 0.5 (0.52, 0.53, 0.51), where before and after the correlation is not significant and much lower.

5°N shows no such correlation increase / change.

(NAO & MHT anomalies both: quarterly, deseasonalized, NOT detrended as before)



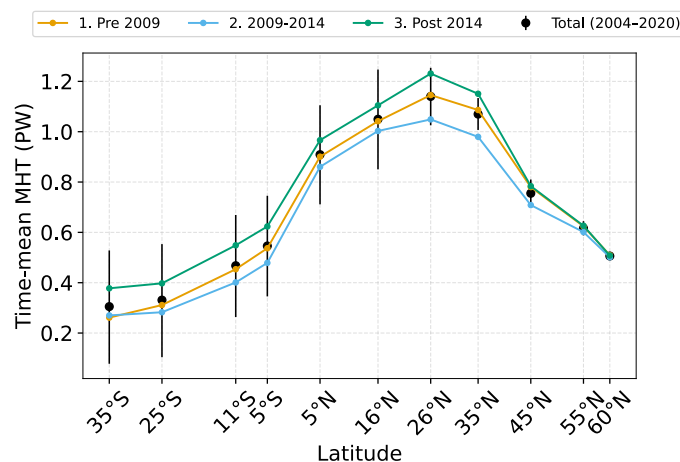
10. When decomposing the heat budget in this region, I find an increase in OHCT, changing from slightly cooling to slightly warming (delta OHCT = +0.017 PW) which is mainly due to 15% higher heat flux uptake in the region, while the HTC (MHT_in - MHT_out) stays around the same.



(All values in PW)

11. The time mean heat transport for each period clearly shows a decrease in MHT across all latitudes in 2009-2014. After 2014 the MHT recovers, while even surpassing the pre2009 baseline at all latitudes.

The reduction of MHT at 16°N is around 3.7% while at 5°N around 4.5%



(10 and 11 together show that while there is a reduction of MHT, it is equivalent in amount across latitudes, leading to no change in OHCT by HTC changes, but increase in OHCT by increased heat flux uptake, which is not transported northward. (Discussion: circulation related reduction in MHT, gyre circulation dominantes in states of reduced overturning, 16°N still in subtropical NA gyre, while 5°N not))

Remark: the figure 5 should be my main figure for part 1 of my results which aims to answer the: How coherent is MHT across the atlantic. Then the second part's aim would be to investigate why the break happens at 16 and 5. Not really a main figure for this part?

LD das ist kein Problem,
denn ich

Appendix

